

SANSKRITI GUPTA

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Education

VIT Bhopal University

B.Tech in Computer Science | GPA: 8.56/10

Expected May 2027

Bhopal, India

Dr. Virendra Swarup Education Centre

CISCE Class XII | 91.5%

2023

Kanpur, India

Projects

Companion – AI-Powered Dementia Care Assistant | LangGraph, LangChain, Llama 3.1, BERT, Python Aug. 2025

- Architected a stateful, multi-agent conversational core using **LangGraph** to handle complex emotional support trees, cyclic dialogue states, and persistent memory retention tailored for senior care.
- Developed a multi-lingual semantic routing pipeline via **LangChain** and **BERT** to classify user intent in real time, separating urgent task-oriented queries from open-ended emotional support interactions.
- Built a privacy-first safety and monitoring system leveraging **Gemini Pro Vision, ViT, and accelerometer-based fall detection** for on-device facial recognition, emergency SOS broadcasting, and GPS-synced caregiver alerts.

TaskGenie – Hyper-Local Service Marketplace | Node.js, PostgreSQL, React, Razorpay, OAuth 2.0 Jun. 2025

- Built a **production-grade hyper-local marketplace** with a **Razorpay-integrated payment gateway** supporting automated refund logic, tiered cancellation policies, and **HMAC SHA256 signature verification** for tamper-proof transactions.
- Engineered **geolocation-based matching APIs** backed by **PostgreSQL B-Tree Indexes and ENUM-typed columns**, achieving a measurable **30% query performance improvement** under concurrent load.
- Architected a **Digital Handshake & Identity system** combining **Google OAuth 2.0** and **4-digit OTP verification** to secure user sessions, validate service completion, and trigger **automated provider payouts with reliability scoring**.

CineBuzz – Movie Discovery & Recommendation Engine | React, REST APIs, Postman Oct. 2024

- Constructed a **dynamic movie discovery platform** indexing **10,000+ titles** with advanced search, genre-based browsing, and trending dashboards achieving a **30% improvement in navigation speed** through strategic UI optimization.
- Designed a **content-based filtering algorithm** that analyses metadata and watch history to generate personalized recommendation lists, improving content relevance for returning users.
- Implemented **client-side caching** to eliminate redundant API calls and validated all integrations via **Postman**, resulting in a measurably faster and more reliable user experience.

Technical Skills

Languages: JavaScript, Java, Python, C++, SQL

Front-End: React.js, Next.js, Flutter, HTML5, CSS3, Redux

Back-End: Node.js, Express.js, Flask, Firebase (Firestore)

Databases: PostgreSQL, MongoDB

AI / GenAI / ML: LangGraph, LangChain, BERT, ViT, Hugging Face

Cloud & DevOps: AWS, Docker, GitHub Actions (CI/CD), Firebase Hosting, Vercel

Tools: Git, GitHub, Postman, Tableau

Leadership & Activities

Google Developers Club – VIT Bhopal

Dec. 2024 – July 2025

Core Member | Web Development, REST APIs, Technical Writing

Bhopal, India

- Conceptualized and led a **Web Development & AI/ML workshop** for **50+ students**, designing a hands-on curriculum that bridged foundational concepts with real-world application.
- Engineered and delivered **RESTful APIs** for a **women's wellness platform**, streamlining data flow across services and improving backend reliability for the club's flagship product.
- Authored "*Agentic AI: Shaping a Future of Autonomous Intelligence*" published on **GDG VIT Bhopal's Medium**, earning significant community engagement and establishing domain thought leadership.

Certifications

AWS Certified Cloud Practitioner – Amazon Web Services

The Bits and Bytes of Computer Networking – Google (Coursera)

Introduction to Machine Learning – NPTEL

Data Analytics and Visualization – Accenture North America Virtual Experience Program